Time-Aware Service Orchestration for Software-Defined Vehicles using TSN and In-Vehicle Networks

Areas

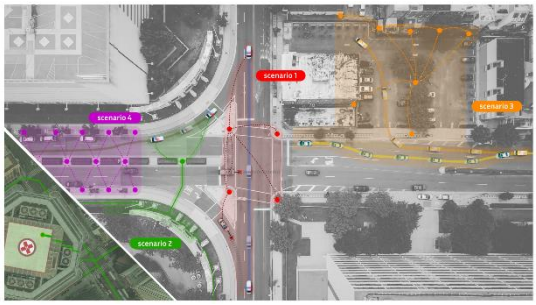
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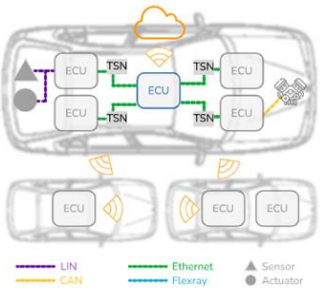
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Motivation

Modern digital infrastructures in the automotive domain are increasingly based on distributed sensing and edge computing within Software-Defined Vehicles (SDVs), where services must be deployed close to in-vehicle data sources to ensure low latency, reliability, and real-time responsiveness. These systems are typically organized into functional zones such as perception, planning, control, and infotainment, each relying on tightly coupled sensors and compute resources interconnected through in-vehicle networks like CAN bus and Ethernet. Time-critical applications such as cooperative perception, sensor fusion, and vehicle control require deterministic execution and communication guarantees. While Time-Sensitive Networking (TSN) provides deterministic behavior at the network level, current container orchestration platforms do not natively support time-awareness or predictable service placement within vehicular environments. This creates a gap between TSN capabilities and application-level orchestration, particularly in autonomous driving stacks such as Autoware running on platforms like PIXKIT.





Objectives

This thesis proposes a time-aware orchestration framework tailored for SDVs, integrating TSN concepts with container orchestration systems such as Kubernetes to enable deterministic service deployment across in-vehicle compute zones. The framework will coordinate with TSN control components such as the Centralized User Configuration (CUC) and Centralized Network Configuration (CNC), ensuring that both compute and network resources are configured according to strict timing requirements. Within the vehicle, the orchestrator will manage services across functional domains (e.g., perception, decision-making, and control), integrating directly with CAN bus data flows and autonomous driving workloads such as Autoware deployed on PIXKIT-based platforms. Application-level requirements, including latency constraints, execution predictability, and service priority, will be translated into coordinated compute placement and TSN network configurations to guarantee deterministic system behavior.

The proposed approach will be evaluated in realistic SDV scenarios using autonomous vehicle platforms, where workloads are distributed across in-vehicle subsystems and must interact reliably with sensor inputs

and control outputs in real time. The focus will be on demonstrating how time-aware orchestration improves predictability, safety, and efficiency in autonomous driving systems by ensuring that services are placed and executed according to strict timing constraints. This work aims to bridge the gap between TSN-enabled deterministic networking and modern SDV orchestration frameworks, enabling more robust and predictable autonomous vehicle architectures.

Work plan

The thesis considers the following activities:

1. State-of-the-art analysis of Time-Sensitive Networks standards (PTP, TAS, CBS, Frame Pre-emption), protocols, and tools for centralized and distributed configuration.
2. Familiarization with the existing TSN equioment, Relyum, and Intel-based.
3. Definition of the architecture, the interaction between components, and components that need to be developed.
4. Implementation of the configuration decisions, using the TSN-based standards available, and the ones developed.
5. Research and implementation of orchestration decisions in the SDV environment.
6. Performance evaluation in the autonomous vehicles and real-world environment, in a smart city testbed, i.e., the Aveiro Tech City Living Lab.
7. Writing the dissertation and a scientific research paper.

The video of the research group is available in https://www.youtube.com/watch?v=dhfJ1wN92nY.
Other videos are available on the following channel https://www.youtube.com/@nap-it.